

ESTIMATION PRO -- HOURS-BASED QUOTATION ENGINE

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

SHA-256: 197d47896b7e2b2b5898efcbf3286ffd8c67c2b42deffe293e4e2e1140aced15

CALCULATION MANIFEST -- Single Source of Truth

Parameter	Value
Piece-Count Subtotal Hrs (Low)	631.46 hrs
Piece-Count Subtotal Hrs (High)	771.81 hrs
Confidence Level	Low
Risk Buffer % Applied	20%
Buffer Hrs Added (Low)	+126.3 hrs
Buffer Hrs Added (High)	+154.4 hrs
LOCK_HRS_LOW (Adjusted Total)	758 hrs
LOCK_HRS_HIGH (Adjusted Total)	926 hrs
Internal Rate	[CONFIDENTIAL]
LOCK_COST_LOW	\$13,600
LOCK_COST_HIGH	\$16,700

> All four LOCK values above are fixed for this entire document. > Sections 1 and 7 must copy these values exactly -- no re-derivation permitted. ---

1. EXECUTIVE SUMMARY

- Project Name / ID: Home 2 - Carowinds Cabelas Dr.
- Total Estimated Hours: 758 - 926 hrs
- Total Estimated Cost (USD): \$13,600 - \$16,700
- Confidence Level: Low
- A significant portion of miscellaneous steel (lintels, bearing plates) had to be estimated as quantities were not provided.
- The full scope of required galvanization is not clearly defined on the drawings, requiring a key assumption.
- Connection details are not fully specified, relying on assumptions of standard AISC shear connections throughout.
- Critical Risks (top 3):
 1. Lintel Quantity Underestimation: Lintel quantities (W-shapes and angles) were estimated. If the actual count from architectural drawings is 50% higher, this could add approximately 85-100 hours to the project.
 2. Undefined Galvanizing Scope: The estimate assumes only loose plates and lintels are galvanized. If all structural steel requires galvanizing, the complexity factor would apply to all beams and columns, adding an estimated 70-90 hours.
 3. Non-Standard Connections: If connections assumed as standard shear tabs are required to be moment, seismic, or otherwise complex, the detailing effort could increase by 40-60 hours depending on the quantity.

2. BASIS OF ESTIMATE

- Drawings Reviewed:
 - S0.01: GENERAL NOTES
 - S0.02: TYPICAL SCHEDULES & DETAILS
 - S0.03: SPECIAL INSPECTIONS
 - S1.01: FOUNDATION PLAN
 - S1.02: SECOND FLOOR FRAMING PLAN
 - S1.03: TYPICAL LEVEL FRAMING PLAN
 - S1.04: ROOF FRAMING PLAN
 - S2.01: STRUCTURAL WALL DESIGNATION PLAN

- S2.02: CMU AND WOOD STUD SCHEDULES
- S2.03: CMU REINFORCEMENT ELEVATIONS - STAIR #1/ELEVATOR
- S2.04: CMU REINFORCEMENT ELEVATIONS - STAIR #2
- S3.01: FOUNDATION SECTIONS
- S3.02: FLOOR FRAMING SECTIONS
- S3.03: ROOF FRAMING SECTIONS
- S3.04: ROOF FRAMING SECTIONS
- S3.05: CANOPY FRAMING
- S3.06: TRELLIS AND SITE WALL DETAILS
- Benchmarks Applied: AISC Manual for standard connection practices, NISD standard hours per piece, and internal historical data for complexity factors.
- Complexity Factor Basis: Multipliers were selected based on explicit information where available (e.g., HSS sections called out) and inferred for others (e.g., CMU interface for lintels and bearing plates). A high-repetition efficiency factor was applied to the main floor beams due to the identical framing on three separate floors (Levels 2, 3, 4).
- Global Assumptions:
 - All beam-to-beam and beam-to-column connections are assumed to be standard single-plate or clip-angle shear connections unless a moment or special connection is explicitly detailed.
 - Steel quantities for three identical floors (Level 2, 3, 4) were derived from sheets S1.02 and S1.03.
 - Galvanizing is assumed to apply only to loose lintels and all bearing plates per notes on S0.02. Main structural frames are assumed to be shop-primed only.

3. ITEMIZED PIECE-COUNT BREAKDOWN

Item Type	Sub-Type	Qty	Base Hrs/Unit	Complexity Factors	Adj Hrs/Unit	Est Hrs (Low)	Est Hrs (High)	Source Sheet	Notes
Columns	Light (HSS 6x6x5/16)	2	3.50	HSS (x1.10)	3.85	6.93	8.47	S1.01	Mark C5

Item Type	Sub-Type	Qty	Base Hrs/Unit	Complexity Factors	Adj Hrs/Unit	Est Hrs (Low)	Est Hrs (High)	Source Sheet	Notes
Columns	Light (HSS 6x6x5/16)	4	3.50	HSS (x1.10)	3.85	13.86	16.94	S1.01	Mark CSA (Wind Post)
Columns	Light (HSS 6x6x3/8)	2	3.50	HSS (x1.10)	3.85	6.93	8.47	S1.01	Mark C6
Columns	Light (HSS 8x6x3/8)	2	3.50	HSS (x1.10)	3.85	6.93	8.47	S1.01	Mark C86
Columns	Light (HSS 8x6x3/8)	Est. 8	3.50	HSS (x1.10)	3.85	27.72	33.88	S3.06	Trellis Columns, Qty Est.
Beams	Simple (<30 ft)	168	2.50	High-rise repetition (x0.85)	2.13	321.48	392.95	S1.02, S1.03	W14x22, 3 floors
Beams	Simple (<30 ft)	3	2.50	Standard complexity (x1.00)	2.50	6.75	8.25	S1.02, S1.03	W16x36, 3 floors
Beams	Standard (30-50 ft)	9	3.50	HSS (x1.10)	3.85	31.19	38.12	S1.02, S1.03	HSS18x6x3/8, 3 floors
Beams	Simple (<30 ft)	3	2.50	HSS (x1.10)	2.75	7.43	9.08	S1.02, S1.03	HSS9x5x3/8, 3 floors
Beams	Simple (<30 ft)	3	2.50	HSS (x1.10)	2.75	7.43	9.08	S1.02, S1.03	HSS6x4x1/4, 3 floors

Item Type	Sub-Type	Qty	Base Hrs/Unit	Complexity Factors	Adj Hrs/Unit	Est Hrs (Low)	Est Hrs (High)	Source Sheet	Notes
Beams	Simple (<30 ft)	Est . 4	2.50	HSS (x1.10)	2.75	9.90	12.10	S3.06	Trellis Beams, Qty Est.
Misc Steel	Plate/embed	Est . 10	0.75	CMU (x1.30), Galv (x1.15) = x1.50	1.13	10.17	12.43	S0.02	Lintel L1, Qty Est.
Misc Steel	Plate/embed	Est . 5	0.75	CMU (x1.30), Galv (x1.15) = x1.50	1.13	5.09	6.22	S0.02	Lintel L2, Qty Est.
Misc Steel	Angle clip	Est . 50	0.25	CMU (x1.30), Galv (x1.15) = x1.50	0.38	17.10	20.90	S0.02	Loose Angle Lintels, Qty Est.
Misc Steel	Plate/embed	Est . 150	0.75	CMU (x1.30), Galv (x1.15) = x1.50	1.13	152.55	186.45	S0.02	Bearing Plates, Qty Est.
SUBTOTAL		424				631.46	771.81		

4. HOURS BY TASK CATEGORY

Task Category	Est Hrs (Low)	Est Hrs (High)	% of Total	Notes
Modeling	126.29	154.36	20%	High repetition increases modeling efficiency.
Shop Drawings / Editing	284.16	347.31	45%	Standard drawing creation for a large number of repetitive pieces.
Checking	94.72	115.77	15%	Standard checking protocol.
Erection Drawings	31.57	38.59	5%	Simplified due to repetitive floor layouts.
RFIs / Revisions	63.15	77.18	10%	Increased allocation due to missing lintel and galvanizing information.
PM / Coordination	31.57	38.59	5%	Standard project management.
SUBTOTAL	631.46	771.81	100%	

5. CONFIDENCE ASSESSMENT

Confidence Level: Low

Checklist (mark each item): ☐ All member quantities explicitly counted on provided drawings. ☐ Connection types fully detailed or called out (not assumed throughout). ☒ Material grades and surface finishes specified for all items. ☒ Drawings are recent (IFC or equivalent), internally coordinated and consistent. ☐ No significant missing, conflicting, or illegible information identified. ☒ All assumptions are reasonable and fully documented.

Criteria met: 3 of 6

Determination rule: 3 or fewer met -> Low

Gaps -- for each unchecked criterion, state: a) What is missing or assumed. b) What assumption was made in Section 3 to handle it. c) Estimated hour impact if the assumption proves wrong.

- Gap 1: Member Quantities Not Explicit

- a) The quantities for all loose lintels (W-shape and Angle) and bearing plates are not available. The drawings refer to architectural plans which were not provided.

- b) Reasonable quantities (10x L1, 5x L2, 50x Angle Lintels, 150x BPs) were estimated based on building size and typical requirements.

- c) If the actual count is 50% higher, the estimate would increase by approximately 85-100 hours. An RFI is required to confirm these counts.

- Gap 2: Connection Types Not Fully Detailed

- a) Most beam-to-column and beam-to-beam connections are shown without a specific detail callout.

- b) Connections were assumed to be standard AISC single-plate or clip-angle shear connections, which is typical for this type of construction.

- c) If moment frames or other complex, non-standard connections are required, the estimate could increase by 40-60 hours.

- Gap 3: Significant Missing Information (Galvanizing Scope)

- a) General Note 11 on S0.01 states steel to be galvanized must meet ASTM A123, but does not specify which members receive this finish. Note 1 on the Bearing Plate schedule (S0.02) specifies plates are galvanized. The scope for main members is ambiguous.

- b) The assumption was made that only loose members (lintels, bearing plates) are galvanized and main frames are prime-painted.

- c) If all structural steel must be hot-dip galvanized, the 15% complexity factor would apply to an additional ~420 hours of base work, adding 70-90 hours to the total.

6. RISK BUFFER AND ADJUSTED HOURS

Row	Low Hrs	High Hrs
Section 3 Subtotal	631.46	771.81
Risk Buffer (20%)	+126.3	+154.4
ADJUSTED TOTAL HOURS	758	926

Confirmation: 758 and 926 match the MANIFEST exactly.

7. COST CONVERSION (USD)

Field	Value
Adjusted Hours (Low)	758 hrs
Adjusted Hours (High)	926 hrs
Blended Hourly Rate	[CONFIDENTIAL]
Project Cost -- Low (USD)	\$13,600
Project Cost -- High (USD)	\$16,700

8. ASSUMPTIONS AND EXCLUSIONS

Key Assumptions (format: [Item] -- assumed [value] because [reason]): - Beam Connections -- assumed standard single-plate shear tabs based on AISC typical practice; no moment symbols or complex connection details are shown on the framing plans. - Lintel Quantities -- assumed quantities for L1, L2, and loose angle lintels because the drawings state to "SEE ARCHITECTURAL DRAWINGS" which were not provided for this estimate. - Bearing Plate Quantities -- assumed a quantity of 150 plates based on the number of beam ends terminating at CMU walls over three floors; an exact count is not possible without final wall layouts. - Galvanizing Scope -- assumed to apply only to loose lintels and bearing plates per specific note on S0.02, as a global requirement for all steel was not specified. - Trellis Steel Quantity -- assumed a quantity of 8 columns and 4 beams for the trellis structures based on typical details and plan indicators; a specific framing plan for the trellis was not provided.

Exclusions (not included in this estimate): - PE stamping or delegated connection design calculations. - 3D MEP coordination and clash detection. - Vendor-furnished items including joists, metal deck, and pre-engineered stairs. - Detailing for pre-fabricated canopies, noted as "BY OTHERS" on sheet S1.02. - Detailing for any wood framing components. - Rebar or concrete reinforcement detailing.

Potential Scope Creep (for PM awareness): - Design changes to floor framing or column locations during the detailing phase. - Addition of miscellaneous supports for Owner-Furnished Equipment (OFE) not shown on the current drawing set. - An RFI confirming the need for all steel to be galvanized would represent a significant scope increase. - A high volume of RFIs may be required to clarify lintel locations and sizes, potentially impacting the schedule.

10. FINAL RECOMMENDATION AND NEXT STEPS

This estimate is rated Low Confidence and should be used for budgetary purposes only until key information is clarified. The primary risks are the unknown quantities of miscellaneous steel and the undefined scope of galvanizing, which could significantly impact the final hours.

The immediate next step is to issue an RFI to the client requesting the architectural drawings to confirm all lintel quantities, types, and locations. A second RFI should request explicit confirmation of the full scope of steel requiring a galvanized finish. A re-estimate will be required upon receipt of this information; a change of more than 25% in the quantity of miscellaneous steel would be a definite trigger.